

Kinematics and integration



1

a $x(t) = 6t^2$

b 3 s

2

a $s(t) = 3t^2 + 10t + 8$

b 133 m

3

a $s(t) = 4t^3 + 15t^2 + 9t - 6$

b 914 m

4

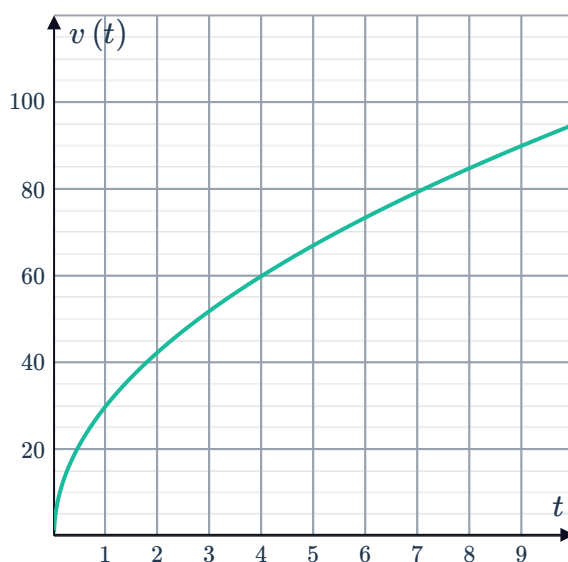
a $x(t) = 8\sqrt{t^3} + 5$

b 221 m to the right of the origin.

5

a 60 m/s

b



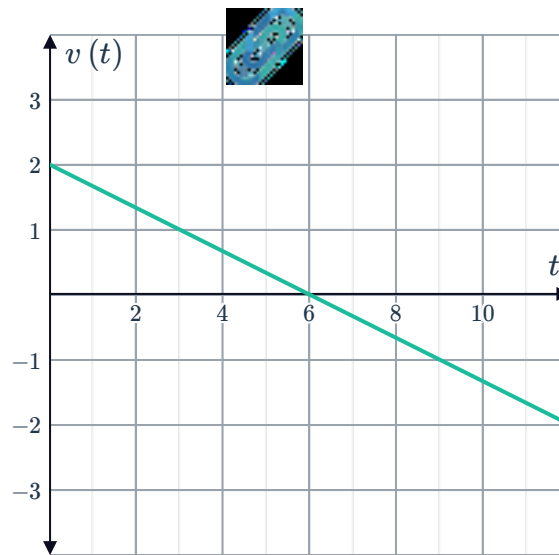
c 160 units²

d The distance travelled by the particle in the first 4 seconds. As the function is positive over this interval the area also represents the displacement of the object from its initial position after 4 seconds.

6

a -1 m/s

b

c 7.5 units^2

d The distance travelled by the particle in the first 9 seconds.

7

a 15 m/s b $x(t) = 3t^2 + 15t + 3$ c 111 m d 66 m

8

a 1 s b $x(t) = 2t^2 - 4t + 3$

c

i 3 ii 1 iii 3

9

a 3 m/s b $x(t) = 2t^3 - 7t^2 + 3t + 2$ c 30 m d 34 m

10

a $x(t) = 4t^3 - 24t^2 + 5$ b $t = 0, 4 \text{ s}$ c -123 m

11

a -2 m/s^2 b $x = \frac{6t^{\frac{5}{2}}}{5} - t^2$ c $t = \frac{4}{9} \text{ s}$ d 2.87 m

12

a $x(t) = t^4 - 12t^3 + 36t^2 + 5$ b A turning point. At $t = 3$ the pen changes direction and starts moving to the left.c 1.96 mg



13 a $5(4 + e^{-5})$ m

b 4.007 m

c 4.007 m

d Yes, because the velocity is always positive for $t \geq 0$.

14 a $v(t) = t^2 - 15t + 56$

b $t = 8, 7$ s

15 a $v(t) = 3t^2 - 27t + 60$

b $t = 4, 5$ s

16 a $v(t) = 2t - 5$

b $x(t) = t^2 - 5t + 8$

c 22 m to the right of the origin.

d $t = 4$ s

17 a $t = 2$ s

b 50 m

c $t = 2 + \sqrt{10}$ s

d $-10\sqrt{10}$ m/s

18 a $t = \frac{75}{49}$ s

b 13.5m

c $t = 3.2$ s

d -16.4 m/s

19 a $v(t) = -\frac{1}{4}e^{4t} + \frac{1}{4}$

b $-30\,322\,823$ cm

20 a 2.98 m/s

b 11.02 m

21 a 278 m/s

b 36 203 m

Trigonometric applications

22 a -18 m/s²

b 3 m

23

a $-9\sqrt{3} \text{ m/s}^2$



b 3 m

24

a $\frac{18}{\pi} \text{ m}$

b $\frac{30}{\pi} \text{ m}$

25

a $\frac{12\sqrt{2}}{\pi} \text{ m}$

b $\frac{96}{\pi} \text{ m}$

26

a $v(t) = -16 \cos 4t + 18$

b 26 m/s

c $x(t) = -4 \sin 4t + 18t + 3$

d $(2\sqrt{3} + 6\pi + 3) \text{ m}$

27

a $v(t) = 10 \sin 5t + 6$

b 11 m/s

c $x(t) = -2 \cos 5t + 6t + 5$

d $(\sqrt{3} + \pi + 5) \text{ m}$

28

a 0 m/s

b $\frac{45}{\pi} \text{ m/s}$

29

a 0 m/s

b $\frac{32}{\pi} \text{ m/s}$